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1. Finding an asymptotically tight bound for $f(n) = 3n^3 - 9$ by using the Θ notation. Justify your answer by demonstrating the constants.
2. Show that $3n^3 - 9 \neq \Omega(n^4)$ by contradiction.

1.

let $f(n) = 3n^3 - 9$, $g(n) = n^3$

$f(n) = \Theta(g(n))$, there exist positive constants c_1, c_2 and n_0 such that

$$c_1 g(n) \leq f(n) \leq c_2 g(n) \quad \forall n \geq n_0$$

$$c_1 n^3 \leq 3n^3 - 9 \leq c_2 n^3 \xrightarrow{\text{同除 } n^3} c_1 \leq 3 - \frac{9}{n^3} \leq c_2$$

$$c_1 \leq 3 - \frac{9}{n^3}$$

$$3 - \frac{9}{n^3} \leq c_2$$

當 $c_1 = 1, c_2 = 4, n_0 = 2$ 時

存在 $n^3 \leq 3n^3 - 9 \leq 4n^3 \quad \forall n \geq 2$

可證明 $3n^3 - 9 = \Theta(n^3)$

$$\begin{aligned} n=1, & \quad c_1 \leq 3 - 9 = -6 \\ n=2, & \quad c_1 \leq 3 - \frac{9}{8} = \frac{15}{8} \\ n=3, & \quad c_1 \leq 3 - \frac{9}{27} = \frac{72}{27} \\ n=4, & \quad c_1 \leq 3 - \frac{9}{64} = \frac{739}{64} \end{aligned}$$

$$\begin{aligned} n=1, & \quad -6 \leq c_2 \\ n=2, & \quad \frac{15}{8} \leq c_2 \\ n=3, & \quad \frac{72}{27} \leq c_2 \\ n=4, & \quad \frac{739}{64} \leq c_2 \end{aligned}$$

隨 n 增長, 當 $n \rightarrow \infty$ 時, $3 - \frac{9}{n^3}$ 之值會趨近於 $3-0$, s.t. $c_1 \leq 3 \leq c_2$

→ 因此, c_1 取 1, c_2 取 4, n_0 取 2

2. if $3n^3 - 9 = \Omega(n^4)$, then there exist positive constants c and n_0 such that

$$cn^4 \leq 3n^3 - 9 \quad \forall n \geq n_0$$

$$cn^4 \leq 3n^3 - 9 \xrightarrow{\text{同除 } n^3} cn \leq 3 - \frac{9}{n^3}$$

只要 $n \geq 4$ 時, 左式必大於 $3 - \frac{9}{n^3}$, 對於假設 $\forall n \geq n_0$ 產生矛盾。

因此, 可證明 $3n^3 - 9 \neq \Omega(n^4)$

目隨 n 增長, 右式在 $n \rightarrow \infty$ 時會趨近於 $3-0=3$